Edward Ngo

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Education

University of Houston December 2025
Bachelors of Science in Computer Science, Minor in Mathematics *Houston, TX*

Experience

Meta January 2026 - Present
Software Engineer *Menlo Park, CA*

Applied AI – Robotics & Control *May 2026 - Present*

- Built a **synthetic agentic pipeline** to generate **280+ RL environments** for **Muse** post-training.
- Designed **230+ robotics scenarios** with **Real2Sim**, **MuJoCo** and **Model Hardware Standard**.
- Developed an **auto-repair agent** that hardens environments with GPT + Opus **pass rate at 10%.**
- Scaled the pipeline to **50** users across the intelligence track, reducing idea-to-environment time to **18 hours**.

Ads Location *January 2026 - May 2026*

- Built an **LLM audit system** that auto-verifies user-location data for **46M+ users** at $\sim$ **89%** accuracy, saving **$600K/year** in annual human review cost.
- Developed a **Llama 3.3 inference pipeline** that evaluates location predictions across **FB/IG/Threads** (**2.8M+ posts/day**), proving **3x** accuracy over the prior model.
- Shipped **ad-targeting location models** on **Spark/SQL** pipelines, updating predictions for **675M+ users** across **3.1B** records.

Meta June 2025 - August 2025
Software Engineer Intern *Menlo Park, CA*

- Integrated **Meta AI Agent** with **Llama 4.0**, reduced **6+ week review cycles** and **53%** late deliveries.
- Built **10+ APIs** with **PHP/GraphQL** to consolidate **15 tables to 1**, reduced **2.8M to 3.3K rows** (**99.9%**).
- Deprecated **70k LOC** by removing legacy routes, improved test coverage **+6.3%** and simplified maintenance.

NASA May 2023 - December 2024
Software Engineer Intern *Houston, TX*

- Developed **on-device LLM platform** with **Llama 3.2** and **Python** for **10+ engineers**, saved **+$100,000**.
- Optimized chatbot **loading time** by **99.4%** with **FAISS** and **CUDA** through a **data saving** feature.
- Implemented **RAG** with **LangChain** and **Transformers**, improved response speed by **40%**.

Projects

AMD $\times$ GPU MODE Hackathon | *HIP, C++, ROCm, MI355X*

- Top **20 (#14)** on the MLA-decode leaderboard (**$1.1M** prize pool), writing custom **HIP** kernels for **FP4 GEMM**, **MoE**, and **MLA-decode**.
- Sped up the **MLA-decode** kernel **1.5 times** (**56.6μs** $\rightarrow$ **36μs**) through skipping the dynamic scale pass and tuning paged attention.
- Optimized an **FP4 quantization** kernel in **HIP**, **3 times** faster than baseline **(12.4μs vs 37.2μs)** at **98.7%** accuracy.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, C #, C++, PHP, Swift, HTML/CSS
Frameworks/Libraries: React, Next.js, Express, .NET, GraphQL, LangChain, FAISS, Tailwind CSS
Tools: PostgreSQL, MySQL, MongoDB, Redis, AWS (EC2/S3), CUDA, Docker, Kubernetes, Linux

Extracurricular

Google: University of Houston Google Developer Group - Campus Lead
Apple: Pathways Alliance Fellow
Awards: NASA CO3 Intern Award, 3x Dean's List, 8x Academic Excellence Scholarship